

JEE MAINS QFT-18 2025



JEE, let's talk about JEE.

Is it anything and everything you have hoped for?

Or do the failure haunt you....

I know the failure haunts you.... ✨



methongo

By: C I P H E R

Ew, I stepped in mathongo



Question Paper

Physics Single Correct (Maximum Marks: 80)

Question No. 1 (Only One Option Correct Type)

The electric field strength depends only on the x , y , and z coordinates according to the law

$$E = \frac{a(x\hat{i} + y\hat{j} + z\hat{k})}{(x^2 + y^2 + z^2)^{3/2}},$$

where $a = 280 \text{ N m}^2 \text{ C}^{-1}$ is a constant. If the potential difference between $(3, 2, 6)$ and $(0, 3, 4)$ is x^2 , what is the value of x ?

- A. 3
- B. 4
- C. 6
- D. 8

Question No. 2 (Only One Option Correct Type)

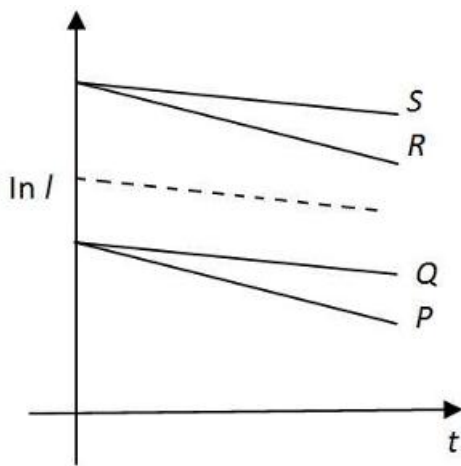
Assertion: The speed of sound in a gas is not affected by a change in pressure provided the temperature of the gas remains constant.

Reason: The speed of sound is inversely proportional to the square root of the density of the gas.

- A. If both assertion and reason are true and reason is the correct explanation of assertion.
- B. If both assertion and reason are true but reason is not the correct explanation of assertion.
- C. If assertion is true but reason is false.
- D. If both assertion and reason are false.

Question No. 3 (Only One Option Correct Type)

A capacitor is charged using an external battery with a resistance x in series. The dashed line shows the variation of $\ln I$ with respect to time. If the resistance is changed to $2x$, the new graph will be:



- (A) P
- (B) R
- (C) S
- (D) Q

Question No. 4 (Only One Option Correct Type)

A thin prism of angle 5° made of glass of refractive index $\mu_1 = 1.5$ is combined with another prism of glass of refractive index $\mu_2 = 1.75$. The combination of the prisms produces dispersion without deviation. The angle of the second prism should be:

- A. 5°
- B. 7°
- C. $\frac{10^\circ}{3}$
- D. 1.2°

Question No. 5 (Only One Option Correct Type)

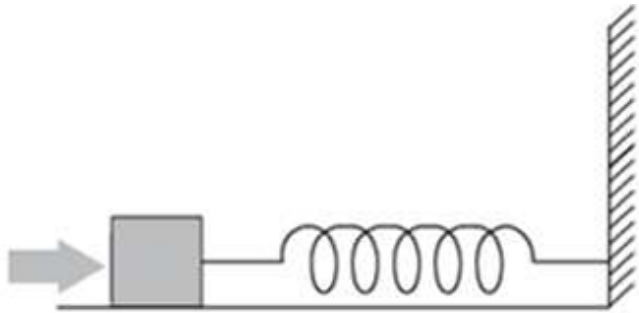
A circular wire loop of radius 1 cm carries a total charge 1×10^{-6} C distributed uniformly over its length. If 0.01% of its length (circumference) is cut off, then the electric field at the center of the loop due to the remaining wire is _____ $\times 10^3$ N/C.

(Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ SI unit)

- A. 6
- B. 5
- C. 8
- D. 9

Question No. 6 (Only One Option Correct Type)

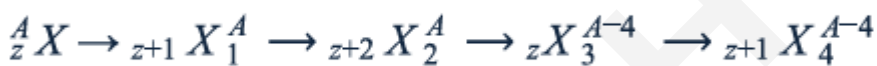
A block of mass 0.18 kg is attached to a spring of force constant 2 N/m . The coefficient of friction between the block and the floor is 0.1 . Initially, the block is at rest, and the spring is unstretched. An impulse is given to the block as shown in the figure. The block slides a distance of 0.06 m and comes to rest for the first time. The initial velocity of the block in m/s is $v = \frac{N}{10}$. Then N is:



- (A) 3
- (B) 2
- (C) 5
- (D) 4

Question No. 7 (Only One Option Correct Type)

In radioactive reaction,

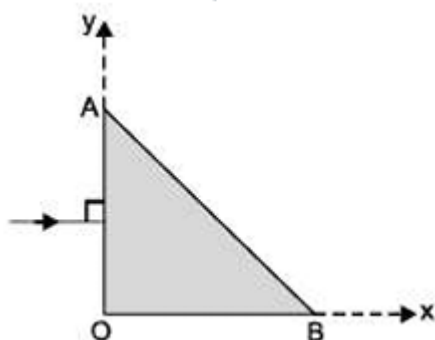


Radioactive radiation are emitted in the sequence:

- A. β^- , β^- , α , α
- B. β^- , β^- , β^+ , α
- C. β^- , β^- , β^- , α
- D. β^- , β^- , α , β^-

Question No. 8 (Only One Option Correct Type)

A triangular medium has a varying refractive index $n = n_0 + ax$, where x is the distance (in cm) along the x -axis from the origin and $n_0 = \frac{4}{3}$. A ray is incident normally on face OA at the midpoint of OA . The range of a so that light does not escape through face AB when it falls first time on the face AB (where $OA = 4 \text{ cm}$, $OB = 3 \text{ cm}$, and $AB = 5 \text{ cm}$) is: (Surrounding medium is air)



A. $a > \frac{1}{9}$

B. $a > \frac{2}{9}$

C. $a > \frac{1}{3}$

D. None of these

Question No. 9 (Only One Option Correct Type)

A Carnot engine with efficiency 50% takes heat from a source at 600 K. To increase the efficiency to 70%, keeping the temperature of the sink the same, the new temperature of the source will be:

A. 360 K

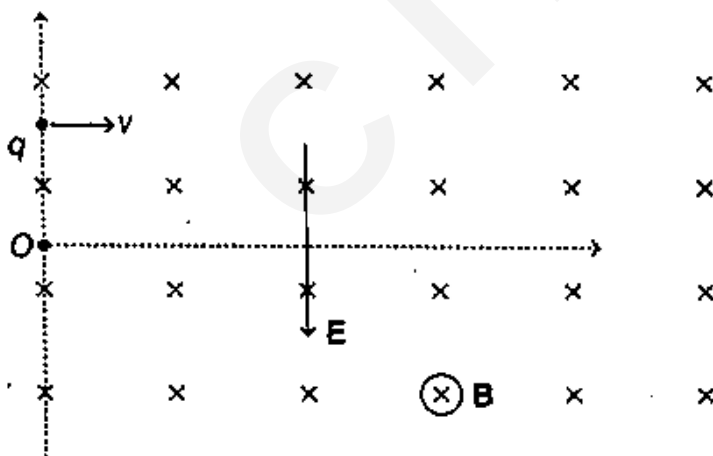
B. 1000 K

C. 900 K

D. 300 K

Question No. 10 (Only One Option Correct Type)

A positively charged particle q of mass m is passed through a velocity selector. It moves horizontally rightward without deviation along the line $y = \frac{2mv}{qB}$ with a speed v . The electric field is vertically downwards, and the magnetic field is into the plane of the paper. Now, the electric field is switched off at $t = 0$. The angular momentum of the charged particle about the origin O at $t = \frac{\pi m}{qB}$ is:



A. $\frac{mE^2}{qB^3}$

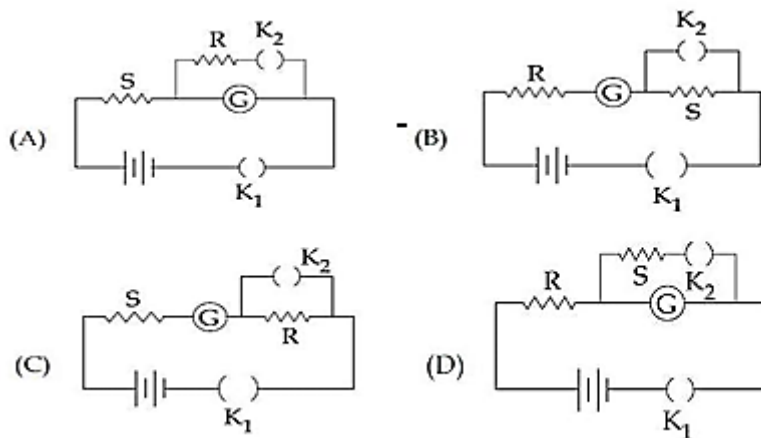
B. $\frac{4m^2E^2}{qB^3}$

C. Zero

D. $\frac{mE^3}{qB^2}$

Question No. 11 (Only One Option Correct Type)

In the circuit diagrams (A, B, C and D) shown below, R is high resistance and S is a resistance of the order of galvanometer resistance G. The correct circuit, corresponding to the half deflection method for finding the resistance and figure of merit of the galvanometer, is the circuit labelled as:



A. Circuit D with $G = \frac{RS}{R-S}$

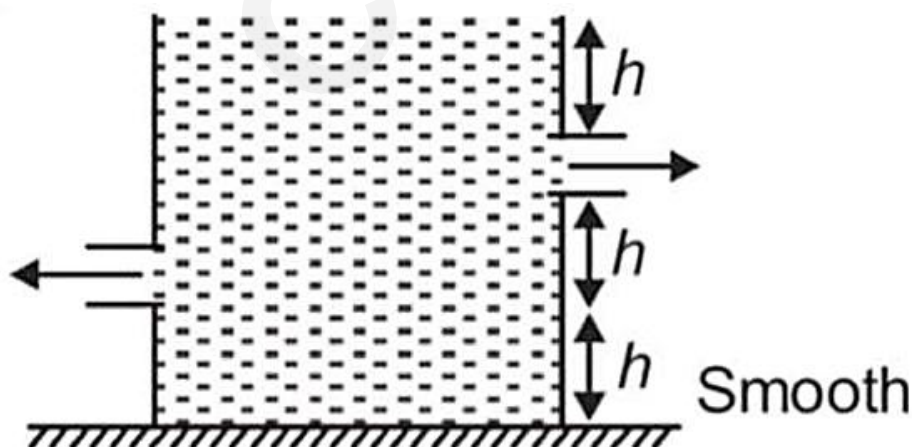
B. Circuit A with $G = \frac{RS}{(R-S)}$

C. Circuit C with $G = S$

D. Circuit B with $G = S$

Question No. 12 (Only One Option Correct Type)

On the opposite sides of a wide vertical vessel filled with water two identical holes are opened, each having cross-sectional area a . If combined mass of water and vessel is M , then find the initial acceleration of tank



A. $\frac{\rho a g h}{M}$

C. $\frac{2 \rho a g h}{M}$

B. $\frac{4 \rho a g h}{M}$

D. Zero

Question No. 13 (Only One Option Correct Type)

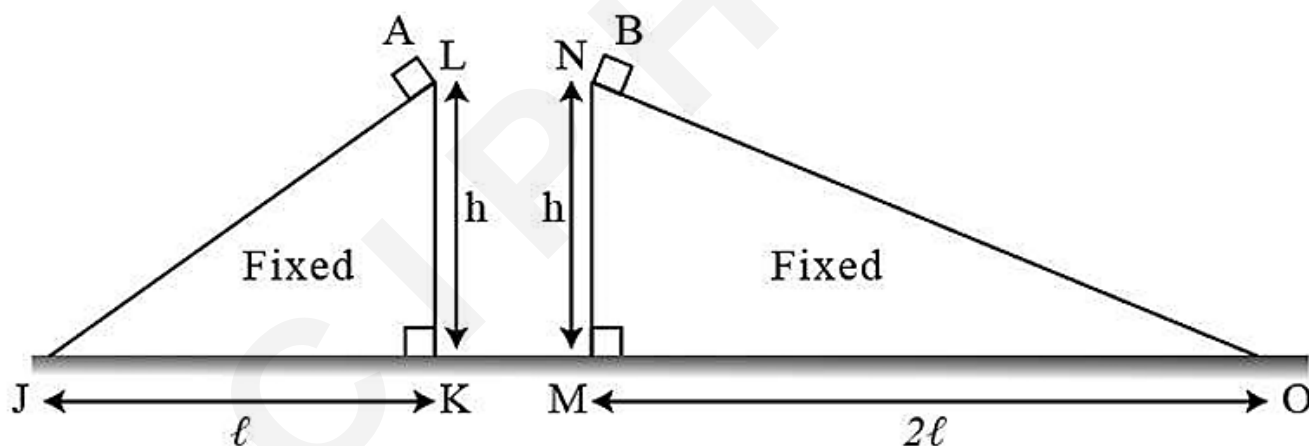
The charge of a parallel plate capacitor is varying as $q = q_0 \sin(2\pi ft)$. The plates are very large and close together (area = A , separation = d). Neglecting the edge effects, the displacement current through the capacitor is:

- A. $\frac{d}{A\epsilon_0}$
- B. $\frac{d}{\epsilon_0} \sin 2\pi ft$
- C. $2\pi f q_0 \cos 2\pi ft$
- D. $\frac{2\pi f q_0}{\epsilon_0} \cos 2\pi ft$

Question No. 14 (Only One Option Correct Type)

Two identical blocks A and B are placed on two inclined planes as shown in the diagram. Neglect air resistance and other friction.

Read the following statements and choose the correct options.



Statements I: Kinetic energy of A on sliding to J will be greater than the kinetic energy of B on falling to M.

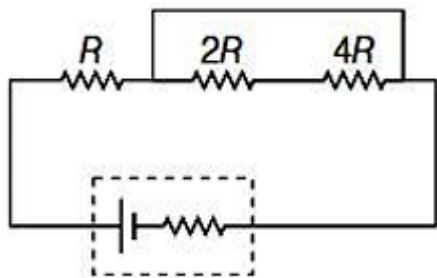
Statements II: Acceleration of A will be greater than the acceleration of B when both are released to slide on the inclined plane.

Statements III: Work done by an external agent to move block slowly from position B to O is negative.

- (A) Only statement I is true.
- (B) Only statement II is true.
- (C) Only I and III are true.
- (D) Only II and III are true.

Question No. 15 (Only One Option Correct Type)

A battery with internal resistance of 4Ω is connected to a circuit consisting three resistances, R , $2R$ and $4R$ (see following figure). If the power generated in the circuit is highest, then the magnitude of R must be



- A. 4Ω
- B. 7Ω
- C. 10Ω
- D. 14Ω

Question No. 16 (Only One Option Correct Type)

Assertion: Machine parts are jammed in winter.

Reason: The viscosity of lubricant used in machine parts increase at low temperatures.

- (A) If both assertion and reason are true but the reason is the correct explanation of assertion.
- (B) If both assertion and reason are true but the reason is not the correct explanation of assertion.
- (C) If assertion is true but reason is false.
- (D) If the assertion and reason both are false.

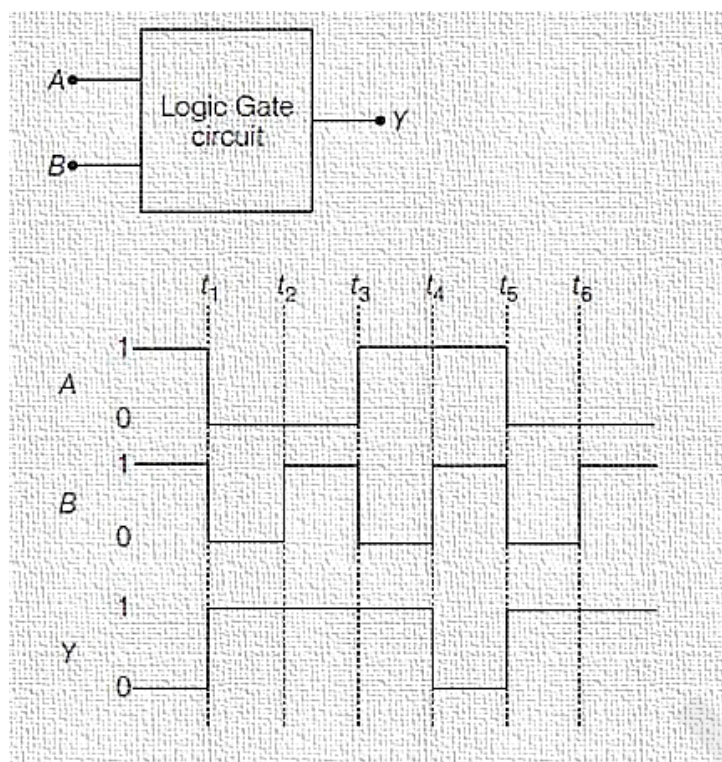
Question No. 17 (Only One Option Correct Type)

A rod of length l has a non-uniform linear mass density given by $\rho(x) = a + b\left(\frac{x}{l}\right)$, where a and b are constants and $0 \leq x \leq l$. The value of x for the center of mass of the rod is at:

- A. $\frac{3}{2} \left(\frac{a+b}{2a+b} \right) l$
- B. $\frac{3}{4} \left(\frac{2a+b}{3a+b} \right) l$
- C. $\frac{4}{3} \left(\frac{a+b}{2a+3b} \right) l$
- D. $\frac{3}{2} \left(\frac{2a+b}{3a+b} \right) l$

Question No. 18 (Only One Option Correct Type)

The following figure shows a logic gate circuit with inputs A and B and output Y . If the voltage waveforms of A , B , and Y are as given, then the logic gate is:



A. NOR gate

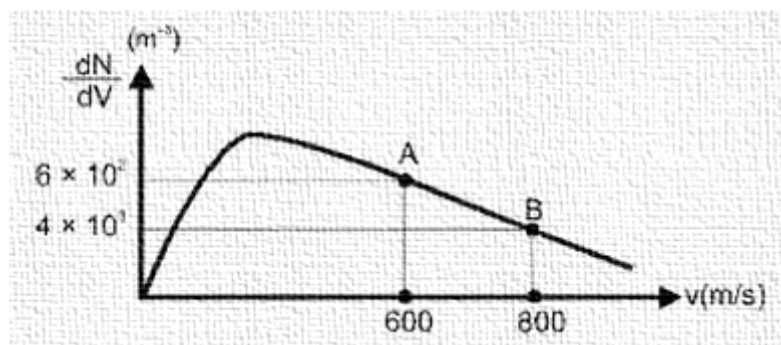
B. OR gate

C. AND gate

D. NAND gate

Question No. 19 (Only One Option Correct Type)

Maxwell's speed distribution curve for an ideal gas is shown in the figure. If the number of molecules in the velocity range from 600 m/s to 800 m/s is 10^n , find the value of n . Assume that the graph between the points A and B is linear.



(A) 3

(B) 6

(C) 5

(D) 2

Question No. 20 (Only One Option Correct Type)

A particle of mass M at rest decays into two masses m_1 and m_2 with non-zero velocities. The ratio of de-Broglie wavelengths of the particles $\frac{\lambda_1}{\lambda_2}$ is:

A. $\frac{m_2}{m_1}$

B. $\frac{m_1}{m_2}$

C. $\frac{\sqrt{m_1}}{\sqrt{m_2}}$

D. 1 : 1

Physics Numerical (Maximum Marks: 20)

Question No. 1 (Numerical Type)

A double-slit arrangement produces interference fringes that are 0.25° apart using wavelength 600 nm. If the angular separation is increased to 0.3° by changing the wavelength of the light used, then the new wavelength must be (Mark answer in nm)

Question No. 2 (Numerical Type)

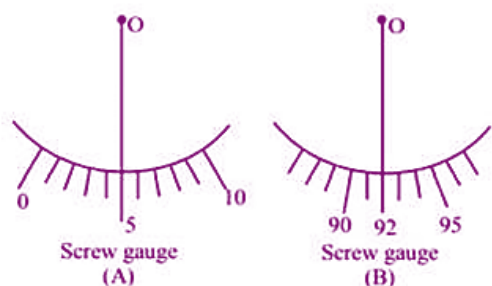
For the equation $F \propto A^a v^b d^c$, where F is the force, A is the area, v is the velocity and d is the density, the value of $3a + 2b + c$ is

Question No. 3 (Numerical Type)

Student A and student B used two screw gauges of equal pitch and 100 equal circular divisions to measure the radius of a given wire. The actual value of the radius of the wire is 0.322 cm. The absolute value of the difference between the final circular scale readings observed by the students A and B is _____.

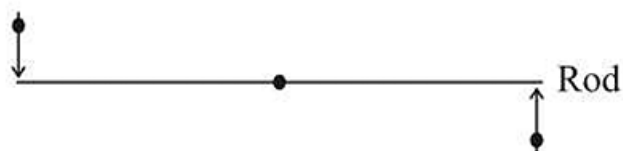
[Figure shows position of reference O when jaws of screw gauge are closed]

Given pitch = 0.1 cm



Question No. 4 (Numerical Type)

A rigid uniform rod of mass M and length L is resting on a smooth horizontal table. It is pivoted at its center. Two marbles, each of mass $\frac{M}{6}$, moving with uniform speed L m/s in the plane of the table, collide with the two ends of the rod simultaneously as shown in the figure. The marbles stick to the rod and continue to move with the rod. The time taken by the rod to rotate through an angle $\frac{\pi}{2}$ radians is $\frac{\pi}{K}$ (in seconds). Find K :



Question No. 5 (Numerical Type)

A solid sphere of mass M and radius R is divided into two unequal parts. The first part has a mass of $\frac{7M}{8}$ and is converted into a uniform disc of radius $2R$. The second part is converted into a uniform solid sphere. Let I_1 be the moment of inertia of the disc about its axis, and I_2 be the moment of inertia of the new sphere about its axis. The ratio $\frac{I_1}{I_2}$ is given by:

Chemistry Single Correct (Maximum Marks: 80)

Question No. 1 (Only One Option Correct Type)

The reagents with which glucose does not react to give the corresponding tests/products are:

- A. Tollen's reagent
- B. Schiff's reagent
- C. HCN
- D. NH_2OH
- E. NaHSO_3

Choose the correct options from the given below:

A. A and D

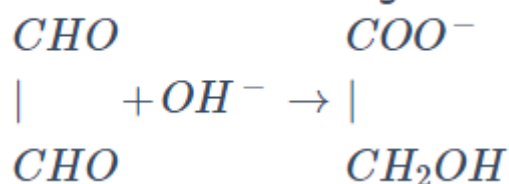
B. B and E

C. E and D

D. B and C

Question No. 2 (Only One Option Correct Type)

Consider the following reaction,



Select the incorrect statement.

A. It is not a disproportionation reaction.

B. It is intramolecular redox reaction.

C. OH^- is a reducing as well as oxidising agent.

D. $\begin{array}{c} \text{CHO} \\ | \end{array}$ is a reducing as well as oxidising agent.

Question No. 3 (Only One Option Correct Type)

When compound X is oxidized by acidified potassium dichromate, compound Y is formed. Compound Y on reduction with LiAlH_4 gives X . X and Y respectively are:

A. $\text{C}_2\text{H}_5\text{OH}$, CH_3COOH

B. CH_3COCH_3 , CH_3COOH

C. $\text{C}_2\text{H}_5\text{OH}$, CH_3COCH_3

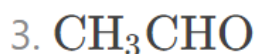
D. CH_3CHO , CH_3COCH_3

Question No. 4 (Only One Option Correct Type)

How many compounds give a positive Tollens' test?

1. D-glucose

2. D-fructose



(A) 3

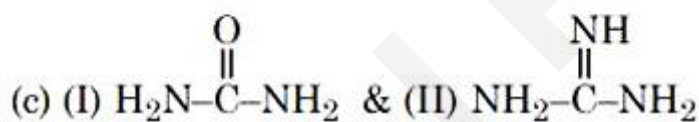
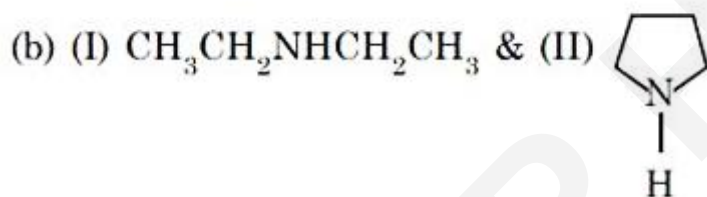
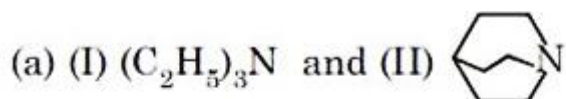
(B) 4

(C) 5

(D) 6

Question No. 5 (Only One Option Correct Type)

More basic between the following pairs of compounds?



A. (a) - (I), (b) - (II), (c) - (I)

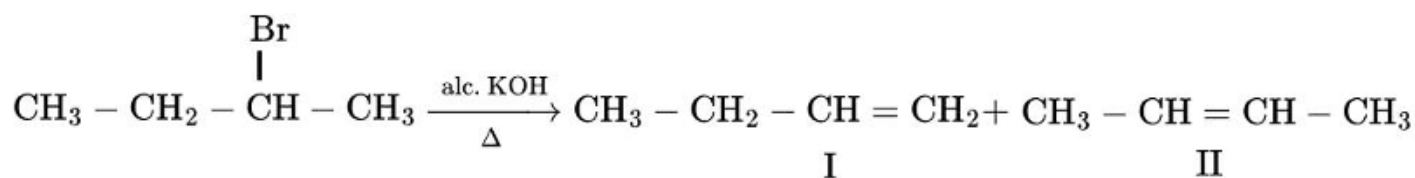
B. (a) - (I), (b) - (I), (c) - (I)

C. (a) - (II), (b) - (I), (c) - (II)

D. (a) - (II), (b) - (II), (c) - (II)

Question No. 6 (Only One Option Correct Type)

Consider the following reaction



- a. I is the major product of the reaction
- b. II is the major product of the reaction
- c. Formation of I is in accordance with Saytzeff's rule
- d. II is more stable because it is more substituted

A. a, c

B. b, c

C. a, d

D. b, d

Question No. 7 (Only One Option Correct Type)

Which of the following statements are correct?

- 1. Electron density in the xy plane in $3d_{x^2-y^2}$ orbital is zero.
- 2. Electron density in the xy plane in $3d_{z^2}$ orbital is zero.
- 3. $2s$ orbital has only one spherical node in it.
- 4. For $2p_x$ orbital, yz is the nodal plane.

A. 3 and 4

B. All of these

C. 2

D. 1 and 3

Question No. 8 (Only One Option Correct Type)

In general, the dissociation constant of a weak monoprotic acid is numerically equal to the dissociation constant of its conjugate base. The pH of a 0.1 M solution of this acid is:

A. 2.5

B. 3

C. 4

D. 5

Question No. 9 (Only One Option Correct Type)

Consider the molecules CH_4 , NH_3 , and H_2O . Which of the given statements is false?

- A. The H-C-H bond angle in CH_4 , the H-N-H bond angle in NH_3 , and the H-O-H bond angle in H_2O are all greater than 90° .
- B. The H-O-H bond angle in H_2O is larger than the H-C-H bond angle in CH_4 .
- C. The H-O-H bond angle in H_2O is smaller than the H-N-H bond angle in NH_3 .
- D. The H-C-H bond angle in CH_4 is larger than the H-N-H bond angle in NH_3 .

Question No. 10 (Only One Option Correct Type)

Assertion (A): Mg^{2+} and Al^{3+} are isoelectronic, but the magnitude of the ionic radius of Al^{3+} is less than that of Mg^{2+} .

Reason (R): The effective nuclear charge on the outermost electrons in Al^{3+} is greater than that in Mg^{2+} .

- A. A is true, R is true, and R is the correct explanation for A.
- B. A is true, R is true, but R is not the correct explanation for A.
- C. A is true, but R is false.
- D. A is false, but R is true.

Question No. 11 (Only One Option Correct Type)

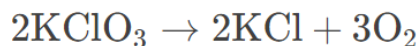
In an acid-catalyzed hydrolysis reaction of ethyl ethanoate, the concentration of the ester changes from 1.50 mol dm^{-3} to 0.35 mol dm^{-3} in 3.5 min, at a given temperature T_1 . Which of the following statements are correct?

- I. The average rate of the reaction is $0.33 \text{ mol dm}^{-3} \text{ min}^{-1}$.
- II. If the reaction is carried out at a higher temperature T_2 , the reaction rate will be greater.
- III. The products of the reaction will be ethanoic acid and ethanol.

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II, and III

Question No. 12 (Only One Option Correct Type)

Which of the following statement(s) is/are not true about the following decomposition reaction?



- (i) Potassium is undergoing oxidation
- (ii) Chlorine is undergoing oxidation
- (iii) Oxygen is reduced
- (iv) None of the species are undergoing oxidation or reduction

A. (i) and (ii)

B. (i) and (iv)

C. (i) and (iii)

D. All of these

Question No. 13 (Only One Option Correct Type)

Statement I: The equivalent weight of redox reactions can be calculated by calculating the change in oxidation numbers. In disproportionation reactions, the equivalent weight can be calculated by separating both redox reactions and adding their equivalent weights.

Statement II: The volume strength of H_2O_2 is a method to describe its concentration. Volume strength is 11.2 times its molarity. A solution of H_2O_2 is given. Its 100 mL is treated with 400 mL $[\text{M}/6]\text{KMnO}_4$ in the presence of H_2SO_4 . The remaining solution is further capable of oxidizing 4.5 g $\text{H}_2\text{C}_2\text{O}_4$.

In statement 2, $\text{H}_2\text{C}_2\text{O}_4$ is converted to CO_2 gas. The volume of this CO_2 gas at N.T.P. is the same as:

A. Oxygen gas formed at NTP by heating of 285.70 gram KClO_3

B. Chlorine gas at NTP by electrolysis of 11.7 gram NaCl (molten)

C. Volume of 1 mole of CO_2 gas at N.T.P.

D. Both (A) and (C)

Question No. 14 (Only One Option Correct Type)

Cold ferrous sulphate solution on absorption of NO develops a brown color due to the formation of:

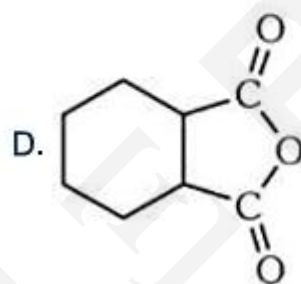
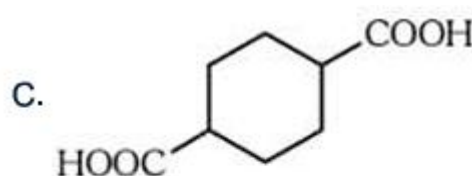
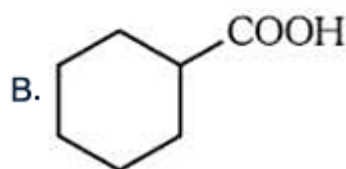
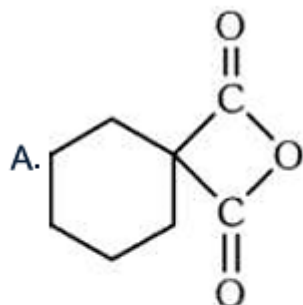
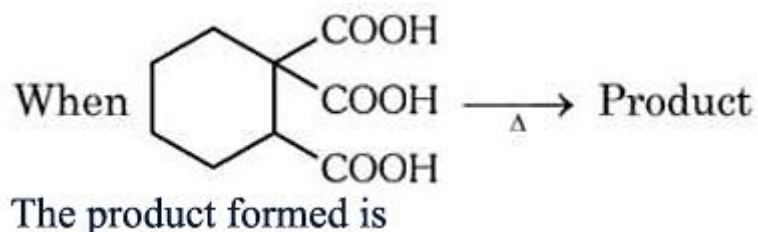
A. Paramagnetic $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]\text{SO}_4$

B. Diamagnetic $[\text{Fe}(\text{H}_2\text{O})_5(\text{N}_3)]\text{SO}_4$

C. Paramagnetic $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO}_3)][\text{SO}_4]_2$

D. Diamagnetic $[\text{Fe}(\text{H}_2\text{O})_4(\text{SO}_4)]\text{NO}_3$

Question No. 15 (Only One Option Correct Type)



Question No. 16 (Only One Option Correct Type)

The freezing point of a solution, prepared from 1.25 g of a non-electrolyte and 20 g of water, is 271.9 K. If the molar depression constant is 1.86 K mol^{-1} , then the molar mass of the solute will be (Round off to the nearest integer):

A. 110

B. 108

C. 106

D. 104

Question No. 17 (Only One Option Correct Type)

Assertion (A): CHCl_3 is stored in closed, dark-colored bottles completely filled, so that air is kept out.

Reason (R): 1° amines on reaction with ethanolic KOH and CHCl_3 give carbylamine.

A. Both A and R are correct, and R is the correct explanation of A.

B. Both A and R are correct, but R is not the correct explanation of A.

C. A is correct, but R is incorrect.

D. Both A and R are incorrect.

Question No. 18 (Only One Option Correct Type)

The formation of the oxide ion, $O^{2-}(g)$, from an oxygen atom requires first an exothermic and then an endothermic step as shown below:



Thus, the process of formation of O^{2-} in the gas phase is unfavorable though O^{2-} is isoelectronic with neon. It is due to the fact that:

A. Electron repulsion outweighs the stability gained by achieving noble gas configuration.

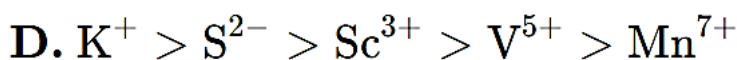
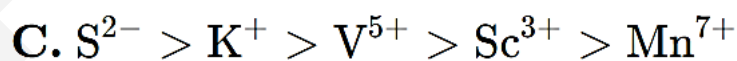
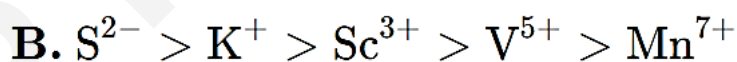
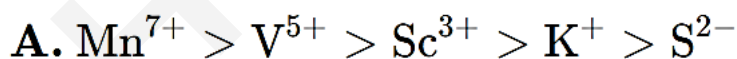
B. O^{-} ion has comparatively smaller size than the oxygen atom.

C. Oxygen is more electronegative.

D. Addition of an electron in oxygen results in a larger size of the ion.

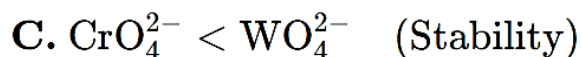
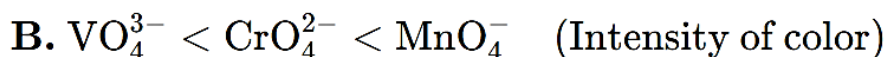
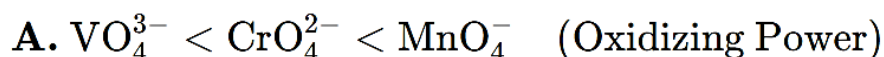
Question No. 19 (Only One Option Correct Type)

Identify the correct order of ionic radii of the given species.



Question No. 20 (Only One Option Correct Type)

Which of the following is correctly matched?



A. A, B only

B. B, C only

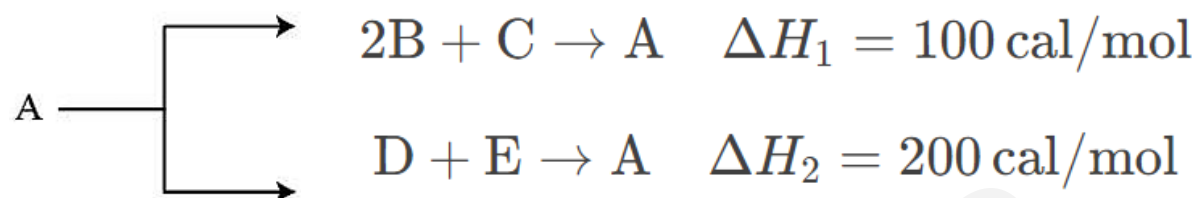
C. A, C only

D. All are correct

Chemistry Numerical (Maximum Marks: 20)

Question No. 1 (Numerical Type)

For the reaction:



If initially 1 mol of A is taken and finally the ratio of B and D is 6 : 1, what is the total enthalpy change (cal) per mole of A reacted?

Question No. 2 (Numerical Type)

5 mL of a gaseous hydrocarbon was exposed to 30 mL of O_2 . The resultant gas, on cooling is found to measure 25 mL, of which 10 mL was absorbed by NaOH and the remainder by pyrogallol. All measurements are made at constant pressure and temperature. The sum of number of carbon and hydrogen atom in the hydrocarbon is

Question No. 3 (Numerical Type)

When CH_3Br and C_2H_5Br are subjected to the Wurtz reaction, the maximum number of possible alkane product(s) formed is/are:

Question No. 4 (Numerical Type)

A crystalline hydrated salt on being heated loses 45.6% of its weight. The percentage composition of the anhydrous salt is Al — 10.5%, K — 15.1%, S — 24.8%, and O — 49.6%. In the molecular formula of the salt (hydrated), how many water molecules are present? (Given: Molar mass of the hydrated salt = 948 g/mol)

Question No. 5 (Numerical Type)

The equivalent conductance of 1 M benzoic acid is $12.8 \Omega^{-1} \text{ cm}^2$, and the conductance of the benzoate ion and H^+ ion at infinite dilution are $42 \Omega^{-1} \text{ cm}^2$ and $288.42 \Omega^{-1} \text{ cm}^2$, respectively. Calculate the percentage of the degree of dissociation. Report your answer by rounding up to the nearest whole number.

Mathematics Single Correct (Maximum Marks: 80)

Question No. 1 (Only One Option Correct Type)

Let the locus of a point z in the Argand plane satisfying the condition $\operatorname{Re}(z^2) = 4$ be C_1 and the locus of z satisfying the condition $\operatorname{Im}(z^2) = 4$ be C_2 . Then the number of common points of the two curves C_1 and C_2 are...

A. 1 B. 2 C. 3 D. 4

Question No. 2 (Only One Option Correct Type)

For $x \in \mathbb{R}$, let $[x]$ denote the greatest integer $\leq x$. Largest natural number n for which

$$E = \left[\frac{\pi}{2} \right] + \left[\frac{1}{100} + \frac{\pi}{2} \right] + \left[\frac{2}{100} + \frac{\pi}{2} \right] + \cdots + \left[\frac{n}{100} + \frac{\pi}{2} \right] < 43$$

is:

A. 39 B. 41 C. 43 D. 37

Question No. 3 (Only One Option Correct Type)

The line $2x + 3y = 6$ cuts x -axis at A and y -axis at B . The line $Kx + 8y = 11$ cuts x -axis at A' and y -axis at B' . If K_1 and K_2 are the values of K such that points A, B, A', B' are concyclic ($K_1 > K_2$), then the value of $\frac{K_1}{4} + \frac{3K_2}{11}$ is...

A. 16 B. 8 C. 2 D. 4

Question No. 4 (Only One Option Correct Type)

If

$$f(x) = \begin{cases} -e^{-x} + k, & x \leq 0 \\ e^x + 1, & 0 < x < 1 \\ ex^2 + \lambda, & x \geq 1 \end{cases}$$

is one-one and monotonically increasing for all $x \in \mathbb{R}$,

then the difference of the maximum value of k and the minimum value of λ is:

A. 0 B. 1 C. 2 D. 3

Question No. 5 (Only One Option Correct Type)

If O, G, S are respectively the orthocentre, centroid, and circumcentre of a triangle whose vertices are $A(2, 3), B(2, 4)$, and $C(4, 3)$, then $AO^2 + 9BG^2 + 4CS^2 =$

A. 11 B. 10 C. 13 D. 12

Question No. 6 (Only One Option Correct Type)

Two parallel chords of a circle of radius 2 are at a distance $\sqrt{3} + 1$ apart. If the chords subtend the angles $\frac{\pi}{k}$ and $\frac{2\pi}{k}$ at the center, where $k > 0$, then the value of $[k]$ is: (Note: $[k]$ denotes the largest integer less than or equal to k)

A. 2 B. 5 C. 4 D. 3

Question No. 7 (Only One Option Correct Type)

Let $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ be a function such that $f((1, 1)) = 2$ and

$$f((m+1, n)) = f((m, n)) + 2(m+n) \quad \text{and} \quad f((m, n+1)) = f((m, n)) + 2(m+n-1),$$

$$\forall m, n \in \mathbb{N}.$$

Then find $f(2, 2)$:

A. 8 B. 12 C. 16 D. 10

Question No. 8 (Only One Option Correct Type)

Let ω be a cube root of unity. If three fair dice are thrown and r_1, r_2 , and r_3 are the numbers obtained on the dice, then the probability that $\omega^{r_1} + \omega^{r_2} + \omega^{r_3} = 0$ is:

A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{2}{9}$ D. $\frac{1}{9}$

Question No. 9 (Only One Option Correct Type)

The lines $\frac{x-4}{15} = \frac{y-17}{9} = \frac{z-11}{8}$ and $\frac{x-15}{4} = \frac{y-9}{17} = \frac{z-8}{11}$ intersect at the point P . Then, the square of the distance of P from the origin is $1400 - \alpha$, where $\alpha =$:

A. 0 B. 4 C. 2 D. 8

Question No. 10 (Only One Option Correct Type)

Let $\cos^{-1}(x) + \cos^{-1}(2x) + \cos^{-1}(3x) = \pi$. If x satisfies the cubic

$$ax^3 + bx^2 + cx - 1 = 0,$$

then $(a + b + c)$ has the value equal to:

A. 24 B. 25 C. 26 D. 27

Question No. 11 (Only One Option Correct Type)

If $A(3, 2, 0)$, $B(5, 3, 2)$, and $C(-9, 6, -3)$ are three points forming a triangle and AD is the bisector of the angle $\angle BAC$, then AD meets BC at the point:

- A. $\left(\frac{19}{8}, \frac{57}{16}, \frac{17}{16}\right)$ B. $\left(-\frac{19}{8}, \frac{57}{16}, \frac{17}{16}\right)$ C. $\left(\frac{19}{8}, -\frac{57}{16}, \frac{17}{16}\right)$ D. none of these

Question No. 12 (Only One Option Correct Type)

The area above the x -axis enclosed by the curves $x^2 - y^2 = 0$ and $x^2 + y - 2 = 0$ is:

- A. $\frac{5}{3}$ B. $\frac{7}{3}$ C. $\frac{8}{3}$ D. $\frac{10}{3}$

Question No. 13 (Only One Option Correct Type)

If α and β are the roots of the equation $x^2 + \alpha x + \beta = 0$ such that $\alpha \neq \beta$, then the number of integral values of x satisfying $|x - \beta| - \alpha < 1$ is:

- A. 0 B. 1 C. 2 D. more than 2

Question No. 14 (Only One Option Correct Type)

Let $\sum_{k=0}^{1000} \binom{2000}{2k} 9^k$ is divisible by 2^a , then the maximum value of a is:

- A. 2001 B. 1998 C. 2000 D. 1999

Question No. 15 (Only One Option Correct Type)

Two numbers are selected at random from the first six positive integers. If X denotes the larger of the two numbers, then $\text{Var}(X) =$:

- A. $\frac{1}{3}$ B. $\frac{14}{3}$ C. $\frac{14}{9}$ D. $\frac{70}{3}$

Question No. 16 (Only One Option Correct Type)

Consider a real-valued continuous function f such that

$$f(x) = \sin x + \int_{-\pi/2}^{\pi/2} (\sin x + tf(t)) dt.$$

Then, the difference of the maximum and minimum value of f in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is:

- A. $\pi + 1$ B. $2(\pi + 1)$ C. π D. can't be determined

Question No. 17 (Only One Option Correct Type)

The value of the Integral:

$$\int_{-\frac{(\pi-2)}{4}}^{\frac{\pi-1}{2}} \lfloor \sin^{-1}(\sin(2x+1)) \rfloor dx$$

(where $\lfloor \cdot \rfloor$ denotes the greatest integer function) is equal to:

A. $\pi - 2$ **B.** 2 **C.** $\frac{\pi}{2} - 1$ **D.** 1

Question No. 18 (**Only One Option Correct Type**)

If all roots of the equation $f(x) = x^6 - 12x^5 + bx^4 + cx^3 + dx^2 + ex + 64 = 0$ are positive, then which has the greatest numerical (absolute) value:

A. b **B.** c **C.** d **D.** e

Question No. 19 (**Only One Option Correct Type**)

Two distinct numbers x and y are chosen at random from the set $\{1, 2, 3, \dots, 30\}$. The number of ways to select these two numbers such that $x^2 - y^2$ is divisible by 3 are:

A. 235 **B.** 240 **C.** 236 **D.** 250

Question No. 20 (**Only One Option Correct Type**)

In a set of real numbers, a relation R is defined as xRy such that $|x| + |y| \leq \frac{1}{2}$. Then, the relation R is:

- A.** reflexive and symmetric but not transitive
- B.** symmetric but not transitive and reflexive
- C.** transitive but not symmetric and reflexive
- D.** none of reflexive, symmetric, and transitive

Mathematics Numerical (Maximum Marks: 20)

Question No. 1 (Numerical Type)

Consider $f(x) = \sin x$ for all $x \in [0, \frac{\pi}{2}]$; $f(x) + f(\pi - x) = 2$ for all $x \in (\frac{\pi}{2}, \pi]$ and

$$f(x) = f(2\pi - x) \quad \text{for all } x \in (\pi, 2\pi).$$

If n, m denote the number of points where $f(x)$ is discontinuous and non-differentiable, respectively, in $[0, 2\pi)$, then the value of $n + m$ is:

Question No. 2 (Numerical Type)

Let A and B be 3×3 matrices with real number entries, where A is symmetric, B is skew-symmetric, and $(A + B)(A - B) = (A - B)(A + B)$. If $(AB)^T = (-1)^k AB$, then the sum of all possible integral values of k in $[2, 10]$ is equal to (where A^T represents the transpose of matrix A):

Question No. 3 (Numerical Type)

Let $a_0 = 0$ and $a_n = 3a_{n-1} + 1$ for $n \geq 1$. Then, the remainder obtained on dividing a_{2010} by 11 is:

Question No. 4 (Numerical Type)

Let P, Q be points of the ellipse $16x^2 + 25y^2 = 400$, so that $PQ = \frac{96}{25}$ and P, Q lie above the major axis. The circle drawn with PQ as diameter touches the major axis at the positive focus. If m is the slope of PQ , then the value of $\frac{1}{|m|}$ is:

Question No. 5 (Numerical Type)

Given the vectors $\vec{a}, \vec{b}, \vec{c}$ such that:

$$\begin{cases} \vec{a} \cdot \vec{c} = 1 \\ \vec{b} \cdot \vec{c} = 4 \\ \vec{b} \times \vec{c} = 2 \\ 2\vec{b} - \vec{c} = \lambda \vec{a} \end{cases}$$

If $\lambda = \sqrt{\alpha - \beta\sqrt{3}}$, where α and β are natural numbers, then the value of $\alpha + \beta$ is:

(ANSWER KEY)

Physics

Q1 []	Q2 []	Q3 []	Q4 []	Q5 []
Q6 []	Q7 []	Q8 []	Q9 []	Q10 []
Q11[]	Q12 []	Q13 []	Q14 []	Q15 []
Q16[]	Q17 []	Q18 []	Q19 []	Q20 []

Q21 []	Q22 []	Q23 []	Q24 []	Q25 []
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Chemistry

Q1 []	Q2 []	Q3 []	Q4 []	Q5 []
Q6 []	Q7 []	Q8 []	Q9 []	Q10 []
Q11[]	Q12 []	Q13 []	Q14 []	Q15 []
Q16[]	Q17 []	Q18 []	Q19 []	Q20 []

Q21 []	Q22 []	Q23 []	Q24 []	Q25 []
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Mathematics

Q1 []	Q2 []	Q3 []	Q4 []	Q5 []
Q6 []	Q7 []	Q8 []	Q9 []	Q10 []
Q11[]	Q12 []	Q13 []	Q14 []	Q15 []
Q16[]	Q17 []	Q18 []	Q19 []	Q20 []

Q21 []	Q22 []	Q23 []	Q24 []	Q25 []
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